Embeddings

A practical reference for representing text and other objects as vectors

1. Definition

An embedding is a dense numerical vector that represents an object such as a word, sentence, document, image, or code fragment in a continuous vector space.

2. Semantic Similarity

Items with related meanings tend to have vectors that are close under a suitable similarity metric. Cosine similarity is commonly used for text embeddings.

3. Typical Pipeline

Text is cleaned and divided into meaningful units, passed through an embedding model, and converted into fixed- or variable-length vector representations. The vectors can then be indexed for retrieval.

4. Vector Databases

Embedding systems are frequently paired with vector indexes to support nearest-neighbor search. This is a key component of retrieval-augmented generation (RAG).

5. Evaluation

Useful checks include retrieval precision, recall, ranking quality, semantic relevance, latency, and robustness to paraphrases.

Concept	Purpose
Embedding model	Transforms an input into a numerical vector.
Cosine similarity	Measures directional similarity between vectors.
Vector index	Efficiently retrieves nearby vectors.
RAG	Uses retrieved information to ground model generation.